

SIMATS ENGINEERING

CLASS TEST

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Course Code:-MLA0401
Course:-Deep Learning

1. Representation Learning in Deep Learning

Introduction

Representation Learning is a technique in which a machine learning model automatically learns useful features or representations from raw data instead of depending completely on manually designed features.

In traditional machine learning, humans usually select important features. In deep learning, neural networks can automatically learn multiple levels of features from the input data.

Example – Image Recognition

Suppose we want to identify whether an image contains a **cat or dog**.

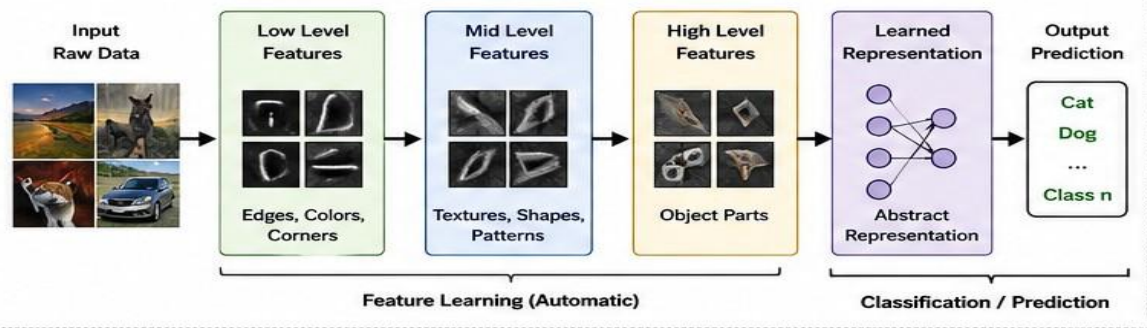
In traditional ML, we may manually extract:

- Color
- Shape
- Edges
- Texture
- Size.

In deep learning, a CNN automatically learns:

- **Lower layers:** edges and corners
- **Middle layers:** shapes, textures, eyes, ears
- **Higher layers:** complete objects such as cats or dogs

1. Representation Learning in Deep Learning



Representation Learning Process

Raw Data → Feature Learning → Representation → Classification/Prediction

For example:

Image → Edges → Shapes → Object Parts → Cat/Dog

Traditional ML vs Representation Learning

Features are often manually designed

Requires feature engineering

Usually shallow models

Works well with smaller datasets

Example: SVM with handcrafted features

Features are often manually designed

Features are learned automatically

Requires less manual feature engineering

Usually deep neural networks

Often benefits from large datasets

Example: CNN for image classification

Features are learned automatically

Advantages

1. Automatically learns useful features.
2. Reduces manual feature engineering.
3. Can learn complex patterns.
4. Useful for images, speech and text.
5. Hierarchical representations improve performance.

Conclusion

Representation learning is one of the major reasons deep learning performs well on complex problems. It allows the network to transform raw data into meaningful representations automatically.

2. Width and Depth of Neural Networks

The **width** and **depth** of a neural network determine its learning capacity and computational complexity.

Width

Width refers to the **number of neurons in a layer**.

For example:

Input

↓

● ● ● ● ● ← 5 neurons

● ● ● ● ●

↓

Output

A wider network contains more neurons and can learn more patterns in the data.

Depth

Depth refers to the **number of layers** in a neural network.

Example:

Input → Hidden Layer 1 → Hidden Layer 2 → Hidden Layer 3 → Output

This network has multiple hidden layers and is therefore deeper.

Effect of Increasing Width

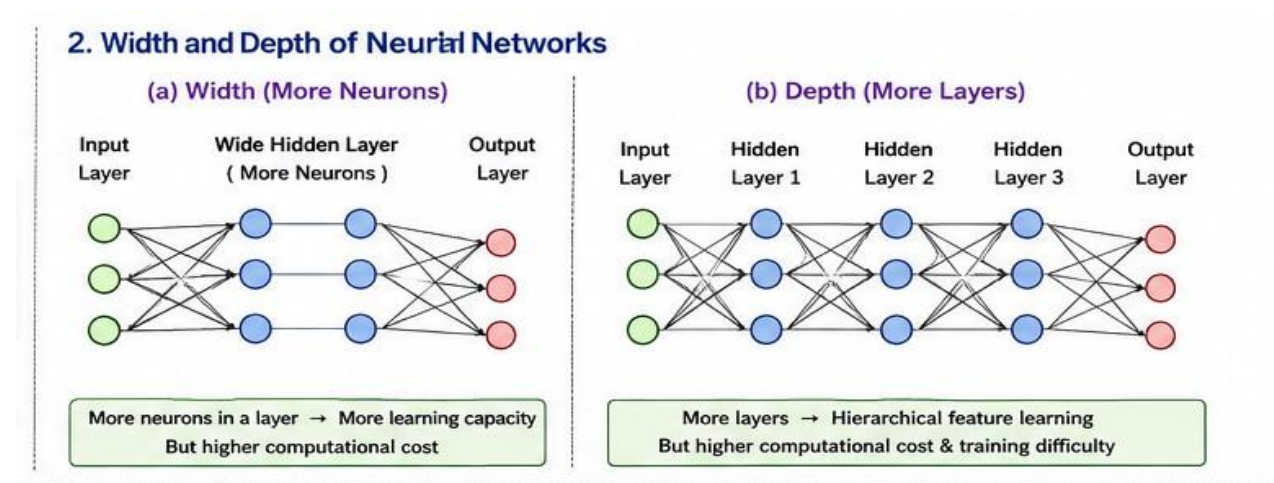
Increasing width:

- Increases the number of parameters.
- Increases learning capacity.
- Allows the network to learn more patterns.
- Increases memory usage and computation.
- May cause overfitting if excessive.

Effect of Increasing Depth

Increasing depth:

- Allows hierarchical feature learning.
- Helps represent complex functions.
- Can improve performance on difficult problems.
- Increases computational complexity.
- Very deep networks may suffer from vanishing/exploding gradients if not properly designed.



Example

For image recognition:

Layer 1 → Edges

Layer 2 → Shapes

Layer 3 → Object parts

Layer 4 → Complete objects

This shows why depth is useful for hierarchical feature learning.

Conclusion

A good neural network requires a suitable balance between width and depth. Increasing them can improve learning capacity, but excessive width or depth can increase computational cost and cause overfitting or training difficulties.

3. ReLU, LReLU and EReLU

Activation functions introduce **non-linearity** into neural networks. This allows neural networks to learn complex relationships.

The three activation functions are **ReLU**, **Leaky ReLU (LReLU)**, and **ELU/EReLU**.

A. ReLU

ReLU stands for **Rectified Linear Unit**.

Mathematical Function

$$f(x) = \max(0, x)$$

Therefore:

- If $x > 0$, output = x
- If $x \leq 0$, output = 0

Example:

$$x = -5 \rightarrow 0$$

$$x = -2 \rightarrow 0$$

$$x = 3 \rightarrow 3$$

$$x = 7 \rightarrow 7$$

Advantages

- Simple and computationally fast.
- Easy to implement.
- Helps reduce the vanishing gradient problem for positive values.
- Widely used in CNNs and deep neural networks.

Limitation

The major problem is the **dying ReLU problem**. Neurons can become inactive when they continuously receive negative values.

B. Leaky ReLU

Leaky ReLU allows a small negative output instead of making all negative values zero.

Mathematical Function

$$f(x) = \begin{cases} x & x \geq 0 \\ \alpha x & x < 0 \end{cases}$$

where α is usually a small value such as 0.01.

Example:

$$x = -5 \rightarrow -0.05$$

$$x = -2 \rightarrow -0.02$$

$$x = 3 \rightarrow 3$$

Advantages

- Reduces the dying ReLU problem.
- Allows gradients for negative inputs.
- Simple and computationally efficient.

Limitation

The negative slope α must be selected appropriately

C. EReLU / ELU

ELU stands for **Exponential Linear Unit**.

Mathematical Function

$$f(x) = \begin{cases} x & x > 0 \\ \alpha(e^x - 1) & x \leq 0 \end{cases}$$

For positive values, it behaves like ReLU. For negative values, it produces a smooth negative output.

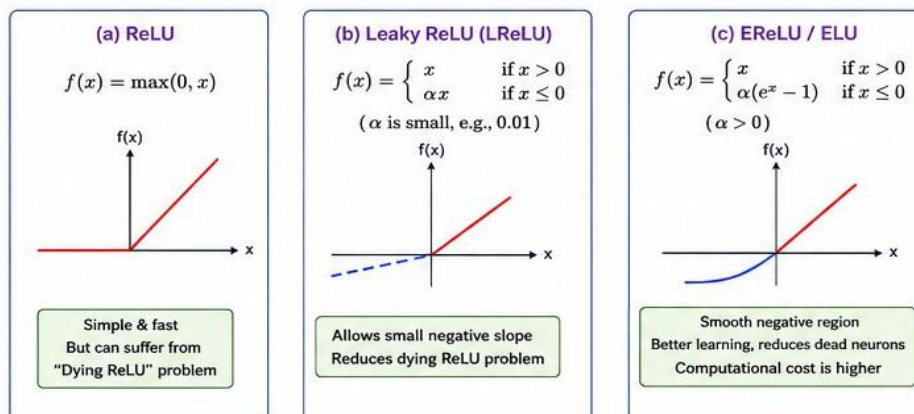
Advantages

- Reduces the dying neuron problem.
- Produces smoother negative outputs.
- Can provide faster and more stable learning in some networks.

Limitation

- More computationally expensive than ReLU.
- Uses exponential calculation.

3. Activation Functions



Conclusion

ReLU is the most commonly used because it is simple and fast. **Leaky ReLU** is useful when avoiding dead neurons is important, while **ELU/EReLU** provides a smoother negative region and can improve optimization in some cases.

4. Unsupervised Training – RBM and Autoencoder

Introduction

Unsupervised learning trains a model using data **without predefined target labels**.

For example, if we have thousands of ECG records without labels such as "normal" or "abnormal", an unsupervised model can learn patterns and representations from the data.

Two important unsupervised neural network techniques are:

1. **Restricted Boltzmann Machine (RBM)**
2. **Autoencoder**

Restricted Boltzmann Machine

An RBM is a generative neural network consisting of two layers:

- **Visible layer**
- **Hidden layer**

There are no connections between neurons within the same layer.

Architecture

Visible Layer

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Hidden Layer

More clearly:

Input/Visible Layer

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Hidden Layer

Working

1. Input data is given to the visible layer.
2. Visible neurons send information to hidden neurons.
3. Hidden neurons learn important patterns.
4. The network attempts to reconstruct the input.
5. Weights are updated to reduce the difference between original and reconstructed data.

RBMs are useful for **feature learning, dimensionality reduction and recommendation systems**.

Autoencoder

An autoencoder is a neural network that learns to reconstruct its input.

It contains three main parts:

Input → Encoder → Latent Representation → Decoder → Output

Architecture

Input Encoder Latent Decoder Output

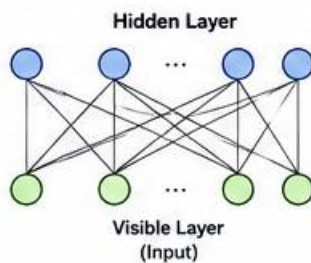


Working

1. Input data is given to the encoder.
2. Encoder compresses the data.
3. A compact representation is created in the latent space.
4. Decoder reconstructs the original input.
5. The model minimizes reconstruction error.

4. Unsupervised Training of Neural Networks

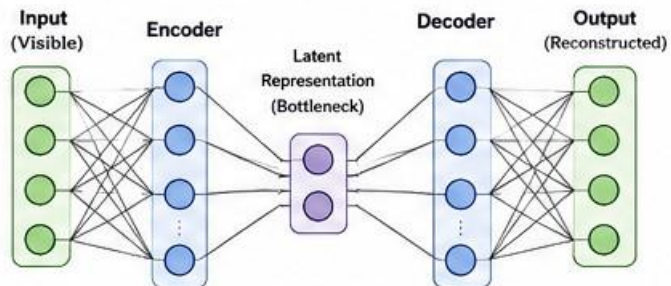
(a) Restricted Boltzmann Machine (RBM)



Working

1. Input given to visible layer
2. Information passes to hidden layer
3. Hidden layer learns patterns
4. Reconstructed back to visible layer
5. Weights updated to reduce difference

(b) Autoencoder



Working

1. Input is encoded into a compact representation (latent space)
2. Decoder reconstructs the original input
3. Model learns to minimize reconstruction error

Conclusion

Both RBMs and autoencoders can learn useful representations without labeled data. Autoencoders are particularly popular for **feature extraction, compression, denoising and anomaly detection**, while RBMs are important probabilistic models for representation and generative learning.

5. Autonomous Systems

Deep learning is an important technology in autonomous vehicles and robots.

It helps systems understand their environment and make decisions.

Applications

- Self-driving cars
- Autonomous robots
- Drones
- Navigation systems
- Obstacle detection

For example:

Camera/Sensors



Environment Detection



Object Recognition



Decision Making



Control

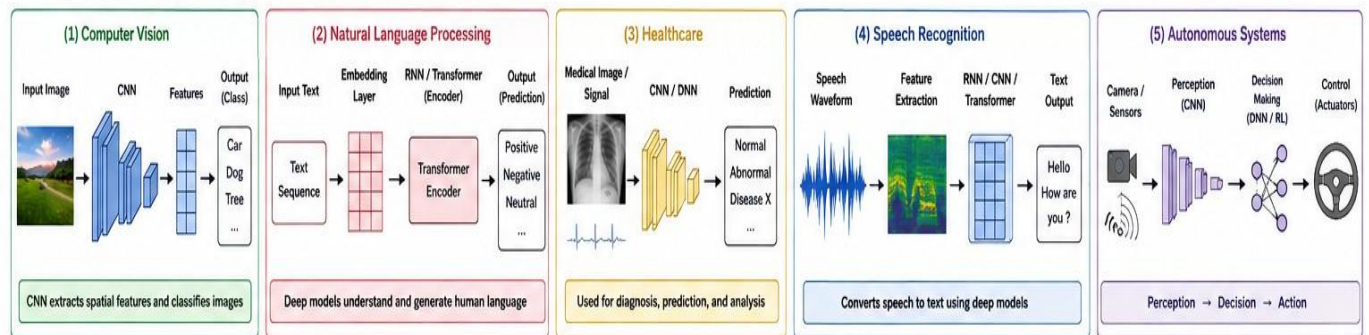
An autonomous vehicle can detect pedestrians, vehicles, traffic signs and road lanes using deep learning.

Other Applications

Deep learning is also used in:

- Recommendation systems
- Fraud detection
- Cybersecurity
- Finance
- Agriculture
- Manufacturing
- Weather prediction
- Generative AI

5. Major Applications of Deep Learning – Architecture Overview



Conclusion

Deep learning has become an important technology in real-world applications because it can automatically learn complex features from large datasets. Its applications range from **computer vision and NLP to healthcare, speech recognition and autonomous systems**, making it one of the most important areas of modern artificial intelligence.

